File permissions in Linux

Project description

You are a security professional at a large organization. You mainly work with their research team. Part of your job is to ensure users on this team are authorized with the appropriate permissions. This helps keep the system secure.

Your task is to examine existing permissions on the file system. You'll need to determine if the permissions match the authorization that should be given. If they do not match, you'll need to modify the permissions to authorize the appropriate users and remove any unauthorized access.

The organization does not allow others to have write access to any files. Based on the permissions established in Step 3, identify which file needs to have its permissions modified. Use a Linux command to modify these permissions.

The research team has archived **.project_x.txt**, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. Use a Linux command to assign **.project_x.txt** the appropriate authorization.

The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents. Use a Linux command to modify the permissions accordingly.

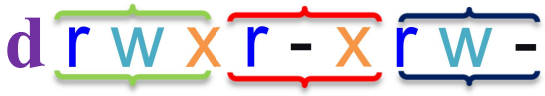
Check file and directory details

Starting with checking the files in directories I wrote the command pwd to check weather I am in the right directory. After determining that I am not in the right directory then used the command cd projects to open the proper directory where I needed to edit permissions. Then I used the commands ls and ls -a to check for file contents as well as check if there are any hidden directory. The results showed that there is one hidden file inside the projects directory, along with 4 .txt files and a drafts directory inside. I then used the ls -la instead of using the ls -l since I determined earlier that there are hidden files inside, this command then displayed the permissions to the files and directories including hidden files.

```
researcher2@bdfcd59afbd9:~$ ls
projects
researcher2@bdfcd59afbd9:~$ pwd
/home/researcher2
researcher2@bdfcd59afbd9:~$ cd projects
researcher2@bdfcd59afbd9:~/projects$ ls
drafts  project_k.txt  project_m.txt  project_r.txt  project_t.txt
researcher2@bdfcd59afbd9:~/projects$ ls -a
.    .project_x.txt  project_k.txt  project_r.txt
..   drafts          project_m.txt  project_t.txt
researcher2@bdfcd59afbd9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 03:24 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 06:12 ..
-rw--w---- 1 researcher2 research_team   46 Sep 13 03:24 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 13 03:24 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Sep 13 03:24 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 13 03:24 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 13 03:24 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 13 03:24 project_t.txt
```

Describe the permissions string

In Linux, permissions are represented with a 10-character string. Permissions include:



1st - character is either a d or a hyphen - this indicates the file type. d indicates that the file is a directory while a - indicates that it is a regular file.

2nd to 4th characters - indicates the permissions for the user in this example. the

2nd character indicates the read permission for user and a - is used when the user lacks read permission.

3rd character indicates the write permission for user and a - is used when the user lacks write permission.

4th character indicates the execute permission for user and a - is used when the user lacks the execute permission.

5th to 7th characters -

5th character indicates the read permission for group and a - is used when the user lacks read permission.

6th character indicates the write permission for group and a - is used when the user lacks write permission.

7th character indicates the execute permission for group and a - is used when the user lacks the execute permission.

8th to 10th characters -

5th character indicates the read permission for other and a - is used when the user lacks read permission.

9th character indicates the write permission for other and a - is used when the user lacks write permission.

10th character indicates the execute permission for other and a - is used when the user lacks the exucute permission.

read: for files, this is the ability to read the file contents; for directories, this is the ability to read all contents in the directory including both files and subdirectories

write: for files, this is the ability to make modifications on the file contents; for directories, this is the ability to create new files in the directory

execute: for files, this is the ability to execute the file if it's a program; for directories, this is the ability to enter the directory and access its files

Change file permissions

To change the permissions I used chmod. The chmod command requires two arguments. The first argument indicates how to change permissions, and the second argument indicates the file or directory that you want to change permissions for. For example the following command would add all permisions to project_m.txt inside our directory:

```
chmod u+rwx,g+rwx,o+rwx project_m.txt
```

If we wanted to take all the permissions away we could use

```
chmod u-rwx,g-rwx,o-rwx project_m.txt
```

Another way to add permission to these groups is to use the equal sign (=). Using = with chmod sets or assigns the permissions exactly as specified. For example lets bring back the original permision for our project_m.txt

```
chmod u=rw, g=r project_m.txt
```

Based on the project descriptions here are the changes that I made.

Since the organization does not allow others to have write access to any files we need to determine which files/directories we need to change.

```
researcher2@36b78a0081e4:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:54 ..
-rw--w---- 1 researcher2 research_team   46 Sep 14 05:16 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 14 05:16 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Sep 14 05:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 14 05:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_t.txt
```

After determining that only project_k.txt is the only file that allows other to write permission I used the command chmod o-w project_k.txt to remove the write permission from others.

```
researcher2@36b78a0081e4:~/projects$ chmod o-w project_k.txt
researcher2@36b78a0081e4:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:54 ..
-rw--w---- 1 researcher2 research_team   46 Sep 14 05:16 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 14 05:16 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 14 05:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_t.txt
researcher2@36b78a0081e4:~/projects$
```

Change file permissions on a hidden file

The research team has archived .project_x.txt, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. In order to change this we use the command chmod u=r, g=r .project_x.txt to change the the file permissions for the hidden file.

Note: Be sure to start the name of a hidden file with a period (.).

```
researcher2@36b78a0081e4:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@36b78a0081e4:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:54 ..
-r--r----- 1 researcher2 research_team   46 Sep 14 05:16 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 14 05:16 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 14 05:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_t.txt
```

Change directory permissions

The files and directories in the projects directory belong to the researcher2 user. Only researcher2 should be allowed to access the drafts directory and its contents. After looking at the permissions set for the `drafts` directory we determined that group also has execute permissions for the folder so we need to change this by using the command `chmod g-x drafts`

```
researcher2@36b78a0081e4:~/projects$ chmod g-x drafts
researcher2@36b78a0081e4:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:16 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 14 05:54 ..
-r--r----- 1 researcher2 research_team   46 Sep 14 05:16 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Sep 14 05:16 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Sep 14 05:16 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Sep 14 05:16 project_t.txt
```

Summary

In this project, I checked the permissions for all the files and directories in the projects folder and changed the ones that did not follow the organization's requirements. This helped ensure that only authorized users in the organization can access or modify the files.